

is important in realising a network protocol for ubiquitous computing is that even general users, who do not have much knowledge of computers, should be able to easily ensure the security of their equipment, confidentiality of stored information and communication, etc. Even now, there are various types of secure protocols, but most of them require operations that need technical knowledge, such as management of authentication and encryption keys, and acquisition of certificates from the certificate authority.

So the researchers at YRP-UNL are working on communication protocols that are simpler and easier-to-use by using tamper-resistant hardware. These protocols will realise a general-purpose security infrastructure that is easy-to-use, packaged, strong and stable. It will enable general users to enjoy the benefits of secure communication infrastructure.

Another challenge in ubi-comp is to truly embed or blend the computers with the environment. Overall design should be aesthetic and calm. The power, noise and other forms of energy consumed and emitted by these devices are also critical as these should not make the users uneasy. Moreover, the computers should mingle, cooperate, support and adjust with the people populating a space, and aid them in their everyday activities. Achieving such characteristics in machines is no mean task, and involves an interdisciplinary approach including biomimetics, artificial intelligence, human-machine interfaces, physics, material sciences and more.

Also, there is a need to develop the required standards to prevent any form of confusion or interoperability issues. "The scope of industry standardisation, especially in using IP as an open standard, has made significant headway. Yet, competing standards and proprietary

protocols persist in many fields. Ubi-comp applications will mean new behaviour and traffic profiles, with architectures very different from those of regular Internet usage. For example, networks will need to cater to lower-energy and smaller nerve-ends, typically operating in more constrained environments. The very broad range of applications and industry sectors likely to benefit from the IoT will demand continued efforts to minimise standardisation issues across industries," says Arora.

Enterprises are the early adopters

Of late, we are seeing sporadic demonstrations of ubiquitous computing. Although many of these are enterprise-scale, there are some country-wide efforts as well.

Arora shares an interesting example to start with: "Many of the existing deployments of ubi-comp have been on *ad-hoc* short-range wireless networks like NFC or low-power and lossy networks. It becomes important to take out the complexity from the spaghetti networks, and make them more intelligent and integrated; the objective being improvement in customer experience and interaction while keeping connectivity costs low. The data collected by smart objects or end-points in a ubiquitous communication system promises to be rich and multi-dimensional, incorporating context, time and location—thus helping to understand behaviour for better customer intimacy.

"A classic example is Coca Cola, which has deployed smart dispensers or end-points, known as Coca Cola Freestyle, in the US to understand and dispense instantly to customer preferences and behaviour. These have a host of other functionalities that include flavour-blending, automatic

refill ordering, content updates, service data collection and anti-vandalism protection measures. Without efficient networking of the end-points, achieving this would not have been possible.

"Platform for last-mile access—be it wired, mobile or fixed wireless—does not matter as long as it meets the requirements (throughput, latency) of the specific end-point.

Globally, we have seen early adoption of ubi-comp to be primarily enterprise-led. This is evident from existing large-scale deployments, such as smart grids for power metering and management, building automation, safety and security, and environmental monitoring. Consumer adoption of the IoT is expected to improve significantly when use-cases become more robust and permeate into mainstream use—possibly driven by business social networking and increased use of smartphones and tablet PCs in enabling location-based consumer intelligence or augmented reality services.

Arora mentions some specific drivers for the growth of ubi-comp in the near future: Improvement of existing applications like RFID, robotics, smartphone and Web tablet (potentially a \$68-billion market worldwide in 2014); new applications in utilities and building automation serving multiple industries, such as transformation of electricity grids into smart grids (a \$76-billion market worldwide in 2014); and new segment solutions and applications like vehicle telematics and smart water management that are expected to grow in the next five years.

Governments doing their bit

Countries like Finland, Japan, China, Malaysia, Singapore and South Korea are doing a lot of multi-disciplinary research



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